SRI LANKA NAVY — DIRECTORATE OF NAVAL CIVIL ENGINEERING

SPECIAL PROJECTS SYSTEM & NAVAL CAPITAL WORKS COMMAND CENTER

CMSys Projects (SP V1 / SPS v4)

Enterprise System Architecture, 6-Cluster Multi-Firebase Topology, 6 Government Vote Heads, 63-File PR Register & Equal Opportunity Rotation

System Platform:	SLN Projects Platform (SPS v4 / SP V1 Command Center)
Core Technology Stack:	ES6+ Native Modules, Reactive Pub-Sub Store (v5), PHP REST Backend
Multi-Firebase Cluster:	6 Distinct Firebase Accounts (Target: cmsys-projects on asia-southeast1)
Naval Capital Works:	25 Major Projects across 6 Government Votes (LKR 448.5M Allocation)
Procurement PR Pipeline:	63 Multi-Stage PR Files (Processing -> Financial Approval -> Received)
Special Projects CSR Suite:	17 Projects: Religious Shrines, Clean Sri Lanka, Married Quarters Housing
Equal Opportunity Engine:	128 Active Sailor Attachments rotated across 661/674 Sailors Pool
Civil PM & QS Studio:	Earned Value Management (EVM), BOQ Costing, Critical Path Gantt (CPM)
Cross-Platform Bridge:	Consolidated JSON Absorption Protocol to CMSys V2 & CE Logistic Panel
Data Security Standard:	Cryptographic Session Token, Prepared Statements, Sanitized UI Binding

CMSys Projects (SP V1) — Strategic Architecture & 6-Cluster Firebase Topology

1. Executive Mandate & Architectural Overview

The Sri Lanka Navy Special Projects System (SP V1 / SPS v4) serves as the primary strategic execution and project engineering platform for naval civil infrastructure. It unifies two critical operational domains: (1) Capital Naval Works funded through national Treasury Vote Heads (Votes 2104, 2001, 2506, 1303, 2103) with a 63-file procurement pipeline, and (2) Special CSR, Welfare, Religious, and Clean Sri Lanka community initiatives governed by an Equal Opportunity Rotation Engine.

To enable real-time collaboration across naval commands without single-point bottlenecks, SP V1 orchestrates a sophisticated 6-cluster Firebase Realtime Database topology paired with an asynchronous absorption bridge that interfaces directly with CMSys V2 and CMSys Logistics.

2. The 6-Cluster Multi-Firebase Account Topology

CLUSTER / PLATFORM	FIREBASE ACCOUNT ID	PRIMARY DATABASE URL	DATA SCOPE & ACCESS MODE
Cluster 1: Projects Platform	cmsys-projects	https://cmsys-projects-default-rtdb.asia-southeast1.firebasedatabase.app	Primary Read/Write: 25 Naval Projects, 63 PP Files, 17 Special Projects, Attachments
Cluster 2: Personnel Master	ce-admin-panel2025	https://ce-admin-panel2025-default-rtdb.firebaseio.com	Federated Read-Only: 661+ Sailors, Trade Skills (MA, CA, PA, PL, WE, BB), Ratings
Cluster 3: Logistics Hub	cmsys-v2-1fb05	https://cmsys-v2-1fb05-default-rtdb.asia...	Target for Absorption Exports: Central stores inventory (1,260 items), NAV/ 254
Cluster 4: Tactical Operations	ncw-ps-operations	https://ncw-ps-operations-default-rtdb.asia...	Operational Peer: Field job cards, daily maintenance work orders, site supervision
Cluster 5: Waterworks Infra	cced-water-distribution	cced-water-distribution-firebaseapp.com (Firestore)	Subsystem Peer: Naval base pipelines, bulk pressure meters, water bowsers
Cluster 6: Archival Fallback	ce-web-system	https://ce-web-system-default-rtdb.asia-southeast1.firebasedatabase.app	Archival Fallback: Historical civil works records, legacy completed project repositories

SYNCHRONIZATION PRINCIPLES & FAILOVER PROTOCOLS

- Dedicated High-Speed RTDB: Cluster 1 (cmsys-projects) isolates heavy project write traffic from daily operations.
- Personnel Federation: Trade quotas are verified against live Cluster 2 sailor availability before attachment approval.
- Zero Data Overwrite: Cluster 3 receives exports in isolated /cmsys_absorption/ nodes to avoid mutating central stores.
- Automatic Offline Fallback: If cloud RTDB drops, client switches transparently to local sample_projects.json cache.

CMSys Projects (SP V1) — Naval Capital Works & Vote Budget Matrix

3. Naval Projects Command Center & 6 Government Vote Heads

The dedicated Naval Projects Command Center oversees 25 major dockyard, coastal engineering, and naval facility construction projects valued at over LKR 448,500,000.00 across 6 distinct financing heads:

GOVERNMENT VOTE HEAD	FINANCING SCOPE	ACTIVE PROJECTS	TOTAL ALLOCATION	COMMITTED FUNDS	EXPENDITURE
Vote 223-1-01-2104	Major New Capital Construction	4 Major Projects	LKR 165,000,000.00	LKR 94,500,000.00	LKR 58,200,000.00
Vote 223-1-01-2001	Major Base & Hospital Renovations	10 Projects	LKR 112,500,000.00	LKR 68,400,000.00	LKR 41,250,000.00
Vote 223-1-01-2506	Maritime Infrastructure & Jetties	4 Projects	LKR 82,000,000.00	LKR 46,800,000.00	LKR 26,400,000.00
Vote 223-1-01-1303	Annual Base Station Maintenance	3 Projects	LKR 28,000,000.00	LKR 14,200,000.00	LKR 8,900,000.00
Non-Naval / Malima Funds	Hospitality & Commercial Works	2 Projects	LKR 35,000,000.00	LKR 18,400,000.00	LKR 9,100,000.00
Vote 223-1-01-2103	Heavy Machinery & Plant Acquisition	2 Projects	LKR 20,000,000.00	LKR 8,000,000.00	LKR 4,400,000.00

4. Flagship Capital Dockyard Engineering Projects

PROJECT CODE	PROJECT TITLE	VOTE NUMBER	BUDGET (LKR)	ACTUAL SPENT	STATUS
NSH	Construction of No.03 Slipway Hangar at Naval Dockyard	Vote 2104	LKR 85,000,000.00	LKR 12,500,000.00	Active
EFS	50m x 20m Steel Workshop Building for Dockyard	Vote 2104	LKR 42,000,000.00	LKR 21,400,000.00	Active
GPE	Guide Pier Extension & Coastal Protection Works	Vote 2506	LKR 26,000,000.00	LKR 14,200,000.00	Active
TNH	Thissa Naval Hospital Two-Story Ward Expansion	Vote 2001	LKR 28,500,000.00	LKR 16,800,000.00	Active
SGP	Front of Beach Security Facility & Command Watchtower	Vote 2104	LKR 18,000,000.00	LKR 9,600,000.00	Active
LMS	Naval Logistics & Management Training School Complex	Vote 2104	LKR 14,000,000.00	LKR 8,500,000.00	Active

CMSys Projects (SP V1) — Master Procurement PR Register (63 Files)

5. The 5-Stage Requisition Pipeline Progression

All material procurement across the 25 naval projects is managed through the Master PR Register, tracking 63 active Purchase Requisitions totaling over LKR 84,500,000.00 across 5 sequential verification stages:

PIPELINE STAGE	ACTIVE FILES	TOTAL STAGE VALUE (LKR)	OPERATIONAL GOVERNANCE & ACTION
1. Processing File	14 Files	LKR 16,420,000.00	Initial bill of quantities vetting, engineering specification check,
2. Awaiting Financial Approval	11 Files	LKR 19,850,000.00	Tender board concurrence, Departmental Tender Board / Ministry
3. Order Placed	18 Files	LKR 26,180,000.00	Official purchase order issued to selected supplier, delivery
4. Items Dispatched	9 Files	LKR 11,250,000.00	Consignment released by vendor, naval transport or commercial
5. Items Received	11 Files	LKR 10,800,000.00	Materials delivered to Dockyard stores, technical inspection passed, taken on-charge.

6. Representative Requisition Line Items

PR NUMBER	PROJECT	COMMODITY DESCRIPTION	QTY	UNIT PRICE	FILE COST (LKR)	STAGE
ENA/CEWS/	NSH	Structural Steel Universal Beams 305x165x40kg/	24 No	LKR 420,000	LKR	Order Placed
ENA/CEWS/	NSH	Portland Composite Cement (PCC) 50kg Bags	1,500 No	LKR 1,850	LKR	Items Received
ENA/CEWS/	EFS	Zinc-Aluminium Roofing Sheets 0.47mm (6m	450 No	LKR 14,200	LKR	Order Placed
ENA/CEWS/	GPE	Tor Steel Reinforcement Bars 20mm High-Tensile	18 Ton	LKR 310,000	LKR	Awaiting
ENA/CEWS/ TNH/2026/0401	TNH	Anti-Microbial Floor Ceramic Tiles 600x600mm	850 Box	LKR 4,600	LKR 3,910,000	Processing File

7. Special Projects CSR & Community Infrastructure Suite

The Special Projects Suite tracks 17 civic, community, and naval welfare initiatives across 4 core categories: (1) Religious Shrines (temples, kovils, churches), (2) Clean Sri Lanka environmental infrastructure, (3) Married Quarters Naval Housing, and (4) Miscellaneous civic assistance.

CATEGORY	PROJECT COUNT	TOTAL BUDGET (LKR)	ACTIVE SAILORS	PRIMARY REGIONS
Religious CSR	6 Projects	LKR 32,500,000.00	48 Sailors	Eastern Command (Trincomalee), North Central (Anuradhapura)
Clean Sri Lanka	4 Projects	LKR 18,200,000.00	26 Sailors	Coastal waste sorting centers, beach conservation nodes, mangrove parks
Naval Housing	5 Projects	LKR 45,000,000.00	42 Sailors	Naval Married Quarters (Peyton Terrace, Vidura, Sandy Bay complexes)
Miscellaneous	2 Projects	LKR 9,800,000.00	12 Sailors	Naval War Memorial refurbishment, sports complex civil maintenance

8. The Equal Opportunity Labour Rotation Engine

The Labour Centre enforces the Headcount Conservation Law and equitable sailor trade rotation across 661 active sailors. It ensures that sailors are not overburdened on out-station projects while preventing skill atrophy in the home yard:

EQUAL OPPORTUNITY ROTATION MATHEMATICAL RULES

- Complement Conservation: Active Duties + Out Projects + Sanctioned Leaves + Camp Standby = 667 (577 VAS / 90 Regular).
- Maximum Attachment Duration: No sailor can exceed 90 continuous days on an out-station project without mandatory rotation.
- Fair Turnaround Buffer: A minimum of 30 days yard standby is enforced between consecutive detachments.
- 6-Trade Matrix Balancing: Strict quotas for Masons (MA), Carpenters (CA), Painters (PA), Plumbers (PL), Welders (WL), Bar Benders (BB).

NAVAL TRADE CODE	TRADE DESCRIPTION	TOTAL POOL	OUT-PROJECT ATTACHED	YARD / STANDBY
MA	Masons (Brickwork, Concrete & Plastering)	142 Sailors	42 Attached	100 On Base / Standby
CA	Carpenters (Joinery, Roof Trusses & Formwork)	98 Sailors	26 Attached	72 On Base / Standby
PA	Painters (Marine Epoxy, Weather-Shield & Spray)	76 Sailors	18 Attached	58 On Base / Standby
PL	Plumbers (Piping, Sanitary & Pressure Pumps)	64 Sailors	16 Attached	48 On Base / Standby
WL / AL	Welders & Aluminum Fabricators	82 Sailors	14 Attached	68 On Base / Standby
BB / RW	Bar Benders & Civil Road Workers	115 Sailors	12 Attached	103 On Base / Standby

9. Quantity Surveying (QS) Studio & Earned Value Management

The Quantity Surveying engine provides mathematical cost and schedule governance for both naval capital and special projects:

EVM METRIC	MATHEMATICAL FORMULA	CURRENT BENCHMARK	INTERPRETATION & ACTION
Planned Value (PV)	PV = BAC * Planned %	LKR 162,400,000.00	Cumulative planned budget expenditure baseline at current date
Earned Value (EV)	EV = BAC * Actual %	LKR 158,800,000.00	Quantified physical civil construction work completed on site
Actual Cost (AC)	AC = Direct Requisitions + Labour	LKR 148,850,000.00	Actual monetary disbursement verified by naval financial ledger
Cost Performance (CPI)	CPI = EV / AC	CPI = 1.067 (Favorable)	Project portfolio is currently operating 6.7% below budgeted cost
Schedule Performance (SPI)	SPI = EV / PV	SPI = 0.978 (Slight Delay)	Project progress is 2.2% behind original milestone schedule.

10. Cross-System CMSys Absorption Protocol

SP V1 acts as an upstream project engineering platform that packages finalized project bills and material requisitions for downstream absorption into CMSys V2 and CMSys Logistics:

CROSS-SYSTEM JSON ABSORPTION SPECIFICATION

- Export Payload Generator: js/cmsys-bridge.js compiles project_meta, work_orders, allocations, and materials bills.
- Logistics Ingestion: CE-Logistic-Panel-v3 consumes the payload via /api/sp-bridge to instantiate official PR files.
- CMSys V2 Headcount Sync: Sailor attachments (128 records) are transmitted to /out_projects to satisfy headcount conservation.
- Store Stock Reservation: Critical materials identified in BOQ are marked as EarmarkedItem to prevent stockouts.

11. Spreadsheet Deliverable Cross-Reference

The complete dataset of 25 naval projects, 63 procurement files, 17 CSR projects, 128 labour attachments, BOQ/WBS items, and the 6-cluster Firebase matrix are compiled into the official multi-worksheet Excel workbook:

MASTER EXCEL SPREADSHEET LEDGER

- File: CMSys_Projects_SPV1_System_Architecture_and_Database_Ledger.xlsx
- Worksheets: 8 Distinct Worksheets (Architecture, Naval Projects, PR Files, Special Projects, Labour, QS BOQ, Job Cards, Bridges)
- Size: 170 KB of verified, fully formatted operational data cross-referenced with this report.